

Ex. 2.1.1 : Find x, if $\begin{vmatrix} x & 2 \\ 8 & 4 \end{vmatrix} = \begin{vmatrix} 1 & 1 \\ 2 & 2 \end{vmatrix}$

(W-07, 2 Marks)

Ex. 2.1.2 : Solve, $\begin{vmatrix} x^2 & -x \\ -5 & 1 \end{vmatrix} = \begin{vmatrix} 7 & -3 \\ 5 & -3 \end{vmatrix}$

(W-09, 2 Marks)

Ex. 2.1.3 : If $A = \begin{vmatrix} -1 & 1 & -1 \\ 2 & 3 & 1 \\ 1 & 2 & 3 \end{vmatrix}$ find $|A|$

(S-07, 2 Marks)

Ex. 2.1.4 : Find x if $\begin{vmatrix} 4 & 3 & 9 \\ 3 & 2 & 7 \\ 1 & 4 & x \end{vmatrix} = 0$

(S-11, 2 Marks)

Ex. 2.1.5 : Find x if $\begin{vmatrix} 1 & 1 & 1 \\ 3 & x & 3 \\ 1 & x & 2 \end{vmatrix} = 0$

(W-16, 2 Marks)

Ex. 2.1.6 : Find x if $\begin{vmatrix} 4 & 3 & 9 \\ 3 & -2 & 7 \\ 11 & 4 & x \end{vmatrix} = 0$

(S-15, W-15, 2 Marks)

Ex. 2.1.7 : Find k if $\begin{vmatrix} 2 & -k & 7 \\ 3 & -4 & 13 \\ 8 & -11 & 33 \end{vmatrix} = 0$

(S-13, 2 Marks)

Ex. 2.1.8 : Solve $\begin{vmatrix} x & 4 & -4 \\ 3 & -2 & 1 \\ -2 & -4 & 1 \end{vmatrix} = 0$

(S-09, 2 Marks)

(e)
$$\begin{vmatrix} 1 & -2 & 4 \\ 1 & x & x^2 \\ 4 & 6 & 9 \end{vmatrix} = \begin{vmatrix} 3 & 6 \\ -2 & -4 \end{vmatrix}$$

Ex. 2.2.1 : Solve by Cramer's rule : $3x + 3y - z = 11$,
 $2x - y + 2z = 9$ and $4x + 3y + 2z = 25$.

(S-10, 4 Marks)

Ex. 2.2.2 : Solve the equation by Cramer's rule

$$x + y + z = 3, \quad x - y + z = 1, \quad x + y - 2z = 0.$$

(S-08, S-12, S-17, S-19, 4 Marks)

Ex. 2.2.3 : Using Cramer's rule solve

$$2x + 3y = 5, y - 3z = -2, z + 3x = 4.$$

(W-09, 4 Marks)

Ex. 2.2.4 : Using Cramer's Rule solve :

$$sx + z = 4, y + z = 2, x + y = 0$$

(W-11, W-12, 2 Marks)

Ex. 2.2.5 : Using determinant method find x , if $x - 2y + 3z = 4$, $2x + y - 3z = 5$, $-x + y + 2z = 3$.

(S-11, 4 Marks)

Ex. 2.2.6 : Using Cramer's Rule solve the equations

$$2x + 4z = 5y + 28 ; x + 11y = 5z - 41 ,$$

$$3x - 3 = 2y + z$$

(S-07, 4 Marks)